

Homework 2: Probability and Relationships

TEMBA Statistics 2026

This assignment covers Chapters 6–10 and 12–14.

Set up the workspace.

1. Unzip IntroStatGPT_homework_02_2026-09-12.zip.
2. Open the ChatGPT desktop app.
3. Choose **File** → **Open Folder**, then select the unzipped TEMBA-2026-homework-02-codex folder.
4. In the task that opens, ask: “Set up this IntroStatGPT workspace and verify that it is ready. Do not begin an analysis yet.”

Do not upload or drag the ZIP file into a chat. Opening the unzipped folder through the desktop app ensures that the workspace instructions and tools load correctly. During setup, IntroStatGPT should check these datasets:

Dataset	Number of rows
fulfillment_shifts	200

Complete the problems. Each **Generate output** panel tells you what to request from IntroStatGPT. Check the resulting tables and plots against those instructions. Write your interpretations in your own words. For hand calculations, show your arithmetic and units. Use a calculator or normal probability function where permitted; follow the specific harness instructions in Problem 2.

Report probabilities to four decimal places and percentages to one decimal place unless a problem specifies otherwise. Keep unrounded values for later calculations. Dollar amounts may be rounded to cents. Label every plot with the variables and their units.

For Problem 2, make one labeled normal-curve sketch and two copies for shading. You may draw digitally or photograph hand-drawn sketches. Most interpretation answers need only two or three sentences.

Assemble and submit your work. After completing the analyses, ask IntroStatGPT to assemble the Problem 2 harness tables and Output A into a Word report in problem order. Check and approve the proposed outline. Then add your numbered answers in Word. Place the three sketches under Problem 2 parts 1–2 and the two harness tables under parts 3 and 5. Export the completed report to PDF.

You may submit one PDF or multiple PDFs. Clearly label every answer with its problem and part number. Scans or photographs of handwritten pages are acceptable if they are legible and converted to PDF. Before submitting, check that your PDF file or files, taken together, contain every numbered answer, Output A, and the sketches, with readable labels. You may refer to labeled values in your tables instead of copying them into your answers.

Problem 1. Evaluate an event-equipment contract

An event-equipment rental company offers a fixed-price weekend contract. Occasionally, a piece of equipment fails during an event and must be replaced immediately. Let X be the number of emergency replacement deliveries required for one contract. The table below is a **supplied hypothetical probability model** for one contract. It is not a table of observed event counts.

Emergency replacement deliveries, x	0	1	2	3
$\Pr(X = x)$	0.57	0.25	0.13	0.05

The company receives \$1,000 for the contract. It incurs \$550 in routine preparation and delivery costs, plus \$175 for each emergency replacement delivery. Profit on one contract is therefore

$$Y = 1,000 - (550 + 175X) = 450 - 175X.$$

1. Calculate the probability of each event:
 - a. At least one emergency replacement delivery.
 - b. At most one emergency replacement delivery.
 - c. The company loses money on the contract.
2. Calculate the expected number of emergency replacement deliveries by multiplying each possible count by its probability and adding. Explain what this average means across many contracts.
3. **Describe profit and its variability.**
 - a. Calculate expected profit using your answer to part 2.
 - b. The number of emergency replacement deliveries has a standard deviation of 0.886. Each delivery reduces profit by \$175. What is the standard deviation of profit?
 - c. Would raising the contract price by \$50 change the standard deviation of profit? Explain.
4. A manager will offer the contract only if expected profit is at least \$325 and the probability of losing money is no more than 0.06. Does this supplied model meet both criteria? Support the decision with your earlier results.

Problem 2. Set an inventory target with a normal demand model

A retailer uses the supplied planning model

$$D \sim N(480, 75^2)$$

for next week's demand for a new product, measured in units. The retailer treats this continuous distribution as an approximation to whole-unit demand. This is a **hypothetical probability model**, not a summary of observed demand.

A **stockout** occurs when demand exceeds the number of units stocked: the retailer runs out before filling all demand for the week.

1. **Sketch the demand distribution.** Draw the normal curve and label its mean. Use the 68/95/99.7 rule to mark the endpoints of the symmetric interval around the mean containing approximately 68% of the probability.
2. **Show the probability regions.** Reproduce your sketch for each event below. Label the relevant demand values and shade the area representing the event. Do not calculate the probabilities yet.
 - a. Demand exceeds 600 units: $\Pr(D > 600)$.
 - b. Demand is between 450 and 600 units: $\Pr(450 < D < 600)$.
3. **Compute the probabilities using the harness.** Ask IntroStatGPT to calculate the two probabilities in part 2 under the supplied normal model. Report each probability to four decimal places and interpret it in one sentence. Check that each result corresponds to the area you shaded.
4. **Set a stockout limit without the harness.** Management will tolerate approximately a 2.5% chance of stockout. Use the 68/95/99.7 rule to determine how many units the retailer should stock. Show your arithmetic. Use a calculator for addition and multiplication, but do not use the harness.
5. **Tighten the limit using the harness.** Management now tolerates a stockout probability of at most 1%. Use IntroStatGPT to find the fewest whole number of units the retailer should stock under the same model. How many more units does this require than your answer to part 4?

Hint for parts 4–5: A stockout happens when demand exceeds stock, so these are upper-tail normal probabilities. If stock is s , the stockout probability is $\Pr(D > s)$. Round the required stock level up to a whole number.

6. **Check the model before relying on it.** The supplied normal QQ plot is a constructed illustration of 80 weeks of demand for a comparable product. Suppose a retailer saw this pattern in its historical data. Would a normal model look reasonable for predicting unusually high demand? Cite a feature of the plot. What would that suggest about relying on the inventory targets in parts 4–5 to meet management's stockout limits?

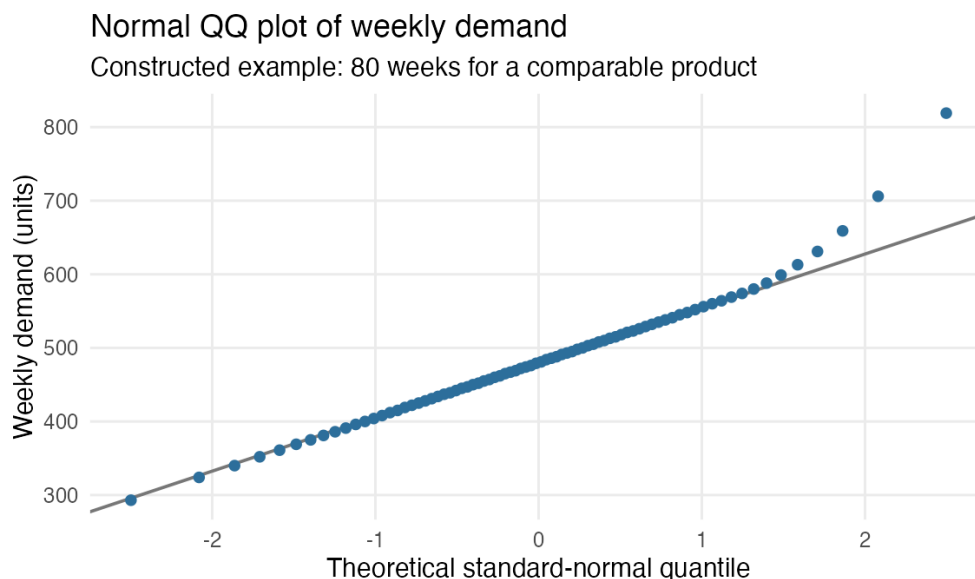


Figure 1: Normal QQ plot for the model check.

Problem 3. Which routing tool resolves tickets faster?

A service center compares its Existing and New ticket-routing tools. The table summarizes 400 tickets, classified as Routine or Complex. “Resolved” means resolved within 24 hours. These counts are **synthetic data** constructed for this assignment.

Case complexity	Existing: resolved	Existing: total tickets	New: resolved	New: total tickets
Routine	135	150	47	50
Complex	20	50	75	150
Total	155	200	122	200

Use the supplied counts for your calculations. Show the numerator and denominator of each percentage.

1. **Compare the tools overall.** Using the Total row, calculate the percentage of tickets resolved within 24 hours for each tool. Which tool looks better?
2. **Compare similar cases.** Calculate each tool’s resolution percentage separately for Routine and Complex tickets: four percentages altogether. Does this comparison support the same conclusion as part 1?
3. **Explain the difference.**
 - a. Calculate three percentages: Routine tickets among Existing’s cases, among New’s cases, and among all tickets.
 - b. Compare these percentages. Are routing tool and case complexity independent in these records? Explain.
 - c. Explain how the different case mixes help account for the conflicting rankings in parts 1–2.

4. **Compare at a common case mix.** Next month, management expects half the tickets to be Routine and half Complex. Assuming each tool maintains its resolution percentage within each category, calculate its expected overall resolution percentage. Which tool would you recommend on this basis? Does this comparison establish that the tool itself causes faster resolution? Explain briefly.

Problem 4. Evaluating a transaction-screening algorithm

A credit card company is evaluating a new algorithm that flags transactions for investigation. Before using it on new transactions, the company runs it on **100,000 historical transactions**.

For this exercise, assume the company knows exactly which historical transactions were fraudulent. Ignore the possibility that some fraud went unnoticed. Running the algorithm produces these counts:

Actual transaction status	Flagged	Not flagged	Total
Fraudulent	900	100	1,000
Not fraudulent	990	98,010	99,000
Total	1,890	98,110	100,000

A flag is a recommendation to investigate, not a determination that fraud occurred.

1. **Express the results as proportions.** Copy the table above and replace every count with its proportion of the **100,000 transactions**. Do this for the row and column totals as well.
2. **Describe the overall results.**
 - a. What percentage of the historical transactions were fraudulent?
 - b. What percentage of all transactions did the algorithm flag?
3. **Check the company's claim.** The company says the algorithm has a **1% false positive rate**: it flags 1% of transactions that are actually legitimate. What percentage of legitimate transactions in this pilot were flagged? Do the pilot results match the claim? Show your numerator and denominator.
4. **Interpret a flag.**
 - a. Among the transactions flagged in the pilot, what percentage were actually fraudulent? Show your numerator and denominator.
 - b. Suppose the company adopts the algorithm and it flags a new transaction. Give these three items:
 - Use your answer to 4(a) to estimate the probability that the new transaction is fraudulent. Write the estimate as a probability statement using Chapter 10's notation.
 - Identify this as a joint, conditional, or marginal probability.
 - State an assumption needed to apply the pilot result to new transactions.
 - c. The salesperson says, "Our algorithm has a 1% false positive rate, so 99% of the transactions it flags are fraudulent." Is that interpretation correct? Explain using the pilot results.
5. **Consider a different credit card product.** Another product has a higher overall percentage of fraudulent transactions. Suppose the algorithm continues to flag **1% of legitimate transactions** and **90% of fraudulent transactions**. Would the percentage of flagged transactions that are actually fraudulent be higher, lower, or unchanged compared with part 4(a)? Explain in words. No formula or numerical calculation is required.

Problem 5. Are busy shifts equally predictable?

A distribution center forecasts when each daily packing shift will finish. The variable `completion_error_minutes` equals the actual completion time minus the forecasted completion time. A negative value means the shift finished early, a positive value means it finished late, and zero means the forecast was exact. Managers treat a forecast as **within target** when its error falls between -10 and 10 minutes, inclusive.

The `fulfillment_shifts` data contain 200 synthetic shifts created for this assignment. Each row represents one shift. `orders_processed` gives the number of orders packed during the shift. The data do not describe a real company.

The planning manager looks at the average completion error and the correlation between workload and completion error. She says, “The forecast stays centered near zero, and the correlation is nearly zero. Our completion forecasts are equally reliable on quiet and busy shifts.”

The supplied figure shows completion-time errors across workloads. The sample correlation between orders processed and completion-time error is -0.0013 .

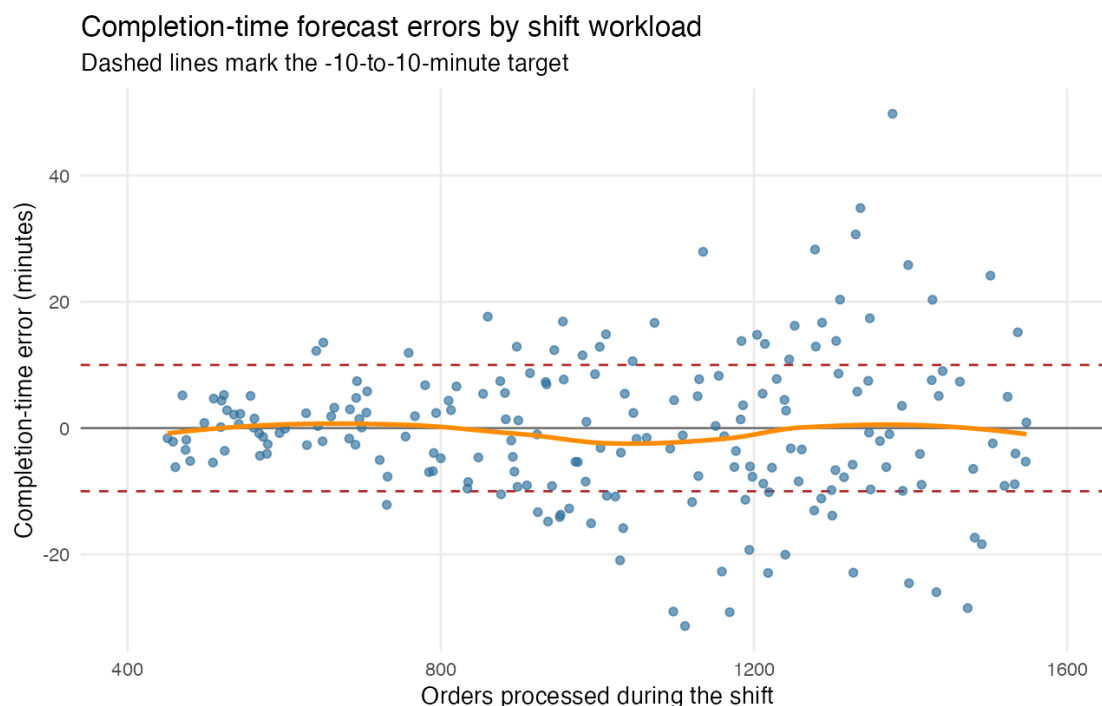


Figure 2: Completion-time errors by shift workload. The smooth curve summarizes the average pattern; dashed lines mark the target limits.

Generate output A: Compare quiet and busy shifts

Use `fulfillment_shifts` to produce the comparison table below.

Comparison table. Divide the shifts into two groups: at most 900 orders, and more than 900 orders. Ask IntroStatGPT to fill in the table below. “Outside target” means an error below -10 minutes or above 10 minutes.

Quantity	At most 900 orders	More than 900 orders
Number of shifts	Fill in.	Fill in.
Mean completion-time error (minutes)	Fill in.	Fill in.
Sample SD of completion-time error (minutes)	Fill in.	Fill in.
Number of shifts outside target	Fill in.	Fill in.
Percentage of shifts outside target	Fill in.	Fill in.

For each group's percentage, divide its number outside target by its number of shifts, then multiply by 100. Give the table a descriptive title.

1. Evaluate the manager's claim.

- Does the average completion-time error change much as workload rises? Support your answer with the supplied smooth curve and correlation and the two group means in your table.
- Are forecasts equally reliable on quiet and busy shifts? Support your answer with the scatterplot's spread and each group's SD and percentage outside target. Explain why the group means and near-zero correlation alone do not establish equal reliability.

Optional harness practice: Reproduce the figure. Use `fulfillment_shifts` to recreate the supplied scatterplot. Put `orders_processed` on the horizontal axis and `completion_error_minutes` on the vertical axis. Add a LOESS smooth curve with span 0.75 and no uncertainty band, a solid horizontal line at zero, and dashed lines at -10 and 10 minutes. Label the axes with their variables and units. Compare your result with the supplied figure; matching its colors and fonts is unnecessary. This practice is not required for submission.